RE-CHECKING THE COLOUR BUDGET OF OPENAI'S PIXTON-COMPLETENESS COUNTEREXAMPLE: GENUS $2.26 \cdot 10^{14}$

[AUTHOR NAME]

Abstract. OpenAI's preprint *A high-arity counterexample to Pixton completeness in Chow* (`openai/math`, family 053) constructs a tautological class Y on $\overline{\mathcal{M}}_{g,n}$, $n = 3\binom{g}{3}$, with $q_{g,n}(Y) = 0$ in CH^* and Y outside Pixton's original relation span, at $g = 10^{60}$. The genus enters the argument only through a finite colour budget (Section 8 of the preprint). Transcribing that budget into exact integer inequalities and optimizing its two free parameters, we find that the same argument closes at $g = 226{,}327{,}553{,}857{,}834 \approx 2.26 \cdot 10^{14}$ (and at $g \approx 9.38 \cdot 10^{13}$ when only the $3K + 5$ colour blocks the preprint proves necessary are reserved). The argument is OpenAI's; we only re-check the budget, assuming the preprint is correct.

1. The budget

Fix g and put $D_* = \binom{g}{3}$, $n = 3D_*$. The preprint chooses integers K and m_0, reserves N_B disjoint blocks of m_0 colours (it takes $N_B = 4K + 20$), and lets S_0 be the remaining colours, so $|S_0| = g - N_B m_0$. Put $C_K = \binom{2K}{3}$, $M_K = \max\{10K + 3, C_K + 1\}$, $e = 2\lfloor m_0/4 \rfloor$, $d_e = \binom{e}{3}$, and $t_{\max} = 25(g-1)^2 - 1 \geq t_b = 25(b-1)^2 - 1$ for every vertex genus $b \leq g$. Section 8 of [1] (`08-support.tex`:326–510), read as exact inequalities, requires:

	condition	source in [1]
P1	$K \geq 5$	$l \geq K$ regime, $b \geq 5$ in Lemma `lem:large-vertex-test`
P2	$g > 2K$	unique large vertex, L370
P3	$4(g - K - 2) > 3g - 3$	at most three nonterminal expensive steps, L455
P4	$3g - 2 < (g - K)^2 - 1$	$\dim S < (b-1)^2 - 1$ with $b \geq g - K + 1$, L421–422
P5	$3gM_K + g(10K + 2) + 2KC_K + t_{\max} \leq d_e/2$	L401–435
P6	$C_K + 1 < 11g$	ordinary cost $\leq 11g$, L470
P7	$\lvert S_0\rvert\big(g(K-4) + 2\big) > 280g^2$	final contradiction, L498–502
P8	$N_B \geq 3K + 5$	block count, L448–458

P5 is what eq. `small-loss` and eq. `large-test-threshold` jointly assert: more than $D_0 - \text{loss} \geq D_0 - d_e/2 + t_b$ internal entries survive, which is the threshold of Lemma `lem:large-vertex-test` (with $t_b \leq d_e/2$). P7 is the exact form of the displayed difference $\frac{|S_0|}{2}(g(l-4)+2) - 140g^2 > 0$ at $l \geq K$; the preprint bounds it via $|S_0| > 0.9g$. The preprint's numerical facts ($C_K < 1.334 \cdot 10^{15}$, $D_0 > 10^9 g^2$, $m_0 \geq 128$, $|S_0| > 0.9g$) serve only to verify P1–P8 at its parameters.

Lemma 1. *Outside Section 8, the proof of the main theorem of* [1] *uses only $g \geq 2$, the identity $n = 3\binom{g}{3}$, and $\deg Y = 2D_* < 3g - 3 + n$, which holds for all $g \geq 2$.*

Justification. We read Sections 1, 2, 7, 8 and searched all sections for `sufficiently large`, `10^{60}`, `g\ge`, `g>`, `D_*`, `3g-3`. The occurrences are $g \geq 2$ (`03-detector.tex`:224, 498, 686; `07-compactification.tex`:132), the degree bound (`02-strata.tex`:235), and one "sufficiently large multiples of K" that concerns a line bundle (`06-coefficient.tex`:474). The genus-b hypotheses of the surface bound are P1, P2, P4. □

Date: 2026-10-09.

Theorem 2. *Assume the proof in* [1] *is correct. Then its main theorem (`thm:main`) and its cohomological corollary hold with*

$$g = 226{,}327{,}553{,}857{,}834, \qquad n = 3\binom{g}{3} \approx 5.80 \cdot 10^{42},$$

using $K = 568$*,* $m_0 = 49{,}723{,}541{,}408$ *and* $4K + 20 = 2292$ *reserved blocks. With* $3K + 5$ *blocks they hold with* $g = 93{,}824{,}084{,}405{,}552$ *(*$K = 567$*,* $m_0 = 27{,}644{,}793{,}412$*). Each value is the least* g *satisfying P1–P8 with the corresponding block rule.*

Proof. At these parameters P1–P8 hold, with the slacks listed in the file `verify_points.txt` of the supplementary scripts; by Lemma 1 nothing else depends on g. For minimality, note that for fixed (K, N_B) a smaller m_0 only helps P7, so we may take $m_0 = 4j$ minimal for P5. P5 then reads $g \le G(j)$ and P2–P4, P6, P7 read $g \ge g_{\min}(j)$, with G of order $j^{3/2}$ and $g_{\min}$ essentially linear in j. The least feasible g is $g_{\min}(j^*)$ for the least j^* with $g_{\min}(j^*) \le G(j^*)$. We computed it by exact integer bisection for every K from 285 (P7 needs $K - 4 > 280$) up to the point where P6 alone exceeds the current optimum, for both block rules. □

Remark 3. The minimum is governed by P5, which forces $m_0 \approx 2(300g^2)^{1/3}$ because $t_b \approx 25g^2$, and by P7, which needs $|S_0| > 280g/(K-4)$. Together they give $g^{1/3} \gtrsim 2 \cdot 300^{1/3} N_B/(1 - 280/(K-4))$, minimized near $K \approx 570$. Improving the genus further would require improving the preprint's bounds themselves, for instance t_b or the $140g^2$ allowance. We do not attempt this.

References

[1] OpenAI, *A high-arity counterexample to Pixton completeness in Chow*, OpenAI Math Release preprint, September 24, 2026, `https://github.com/openai/math` (commit `fd4aeeb2`), directory `preprints/A-high-arity-counterexample-to-Pixton-completeness-in-Chow-September-24-2026`.

[2] A. Pixton, *Conjectural relations in the tautological ring of* $\overline{\mathcal{M}}_{g,n}$, arXiv:1207.1918.

[3] R. Pandharipande, A. Pixton, D. Zvonkine, *Relations on* $\overline{\mathcal{M}}_{g,n}$ *via 3-spin structures*, J. Amer. Math. Soc. 28 (2015), 279–309.

Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.